

Diabetes Detection Using Machine Learning

*A Project Report Submitted in the
Partial Fulfillment of the Requirements
for the Award of the Degree of*

BACHELOR OF TECHNOLOGY

IN

ELECTRONICS AND COMMUNICATION ENGINEERING

Submitted by

G.Rohit	23881A0414
K.Dinesh Reddy	23881A0419
K.Ruchi Rangandha	23881A0420

SUPERVISOR

Dr K H Murali

Assistant Professor,ECE



Department of Electronics and Communication Engineering

June, 2026



Department of Electronics and Communication Engineering

CERTIFICATE

This is to certify that the mini-project titled **Diabetes Detection Using Machine Learning** is carried out by

G.Rohit	23881A0414
K.Dinesh Reddy	23881A0419
K.Ruchi Rangandha	23881A0420

in partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology in Electronics and Communication Engineering** during the year 2025-26.

Signature of the Supervisor
Dr K H Murali
Associate Professor,ECE

Signature of the HOD
Prof.S.Rajendar
Professor and Head, ECE

Acknowledgements

The satisfaction that accompanies the successful completion of the task would be put incomplete without the mention of the people who made it possible, whose constant guidance and encouragement crown all the efforts with success.

We wish to express our deep sense of gratitude to **DR K H Murali**, Assistant Professor,ECE Project Supervisor, Department of Electronics and Communication Engineering, Vardhaman College of Engineering, for his able guidance and useful suggestions, which helped us in completing the project in time.

We are particularly thankful to **Prof.S.Rajendar**, the Head of the Department, Department of Electronics and Communication Engineering, his guidance, intense support and encouragement, which helped us to mould our project into a successful one.

We show gratitude to our honorable Principal **Prof.J.V.R. Ravindra**, for providing all facilities and support.

We avail this opportunity to express our deep sense of gratitude and heartfelt thanks to **Dr. Teegala Vijender Reddy**, Chairman and **Sri Teegala Upender Reddy**, Secretary of VCE, for providing a congenial atmosphere to complete this project successfully.

We also thank all the staff members of Electronics and Communication Engineering department for their valuable support and generous advice. Finally thanks to all our friends and family members for their continuous support and enthusiastic help.

G.Rohit

K.Dinesh Reddy

K.Ruchi Rangandha

Abstract

Diabetes is one of the most common chronic diseases worldwide and has become a major public health concern due to its increasing prevalence and serious health complications. Early detection of diabetes is essential to reduce the risk of severe conditions such as heart disease, kidney failure, and vision impairment. Conventional diagnostic methods mainly rely on laboratory tests and clinical examinations, which may be time-consuming and require medical expertise. This project presents a Diabetes Prediction System using Machine Learning to assist in the early identification of diabetes based on patient health data.

The system utilizes the Pima Indians Diabetes Dataset, which contains medical information such as pregnancies, glucose level, blood pressure, skin thickness, insulin, body mass index (BMI), diabetes pedigree function, and age. Data preprocessing techniques were applied before training the machine learning models. Three classification algorithms, namely Logistic Regression, Decision Tree, and Random Forest, were implemented and evaluated to determine their prediction performance. The models were compared based on prediction accuracy, and the best-performing model was selected for diabetes prediction.

The experimental results showed that Logistic Regression and Decision Tree achieved an accuracy of 74, while Random Forest achieved an accuracy of 72. The developed system provides a simple and efficient approach for predicting diabetes risk using machine learning techniques. This project demonstrates the potential of artificial intelligence in supporting healthcare professionals by enabling early disease prediction and improving decision-making. The proposed system can be further enhanced by using larger datasets, advanced machine learning algorithms, and web-based deployment for real-time healthcare applications.

Keywords: Diabetes Prediction, Machine Learning, Logistic Regression, Decision Tree, Random Forest, Healthcare, Classification.

Table of Contents

Title	Page No.
Acknowledgements	i
Abstract	ii
List of Tables	v
List of Figures	vi
Abbreviations	vi
CHAPTER 1 Introduction	1
1.1 Introduction	1
1.2 Background and Motivation	4
1.2.1 Objectives of the Project	5
1.2.2 Organization of the Project	6
CHAPTER 2 Literature Survey	7
2.1 Introduction	7
2.2 Identified Research Gaps	9
2.3 Summary	9
CHAPTER 3 Review of Prior Research	10
3.1 Introduction	10
3.2 Hardware Implementation Considerations	11
3.3 Summary	11
CHAPTER 4 Proposed Architecture	13
4.1 Introduction	13
4.2 Summary	14
CHAPTER 5 Results and Discussions	16
5.1 Introduction	16
5.2 Simulation Environment	16
5.3 Functional Verification	19
5.4 Discussion	19
5.5 Summary	20
CHAPTER 6 Conclusions and Future Scope	21
6.1 Introduction	21
6.2 Summary of Work	22

6.3	Key Observations	23
6.4	Conclusion	24
CHAPTER 7 Mapping of Program Outcomes (POs) and Sus-		
tainable Development Goals (SDGs)		25
7.1	Overall Impact	27

List of Tables

5.1	Performance Comparison of Machine Learning Algorithms	16
-----	---	----

List of Figures

4.1	Proposed Architecture of the Diabetes Prediction System	14
5.1	Console Output Showing Model Accuracy and Diabetes Prediction Result	17
5.2	Console Output Showing Model Accuracy and Diabetes Prediction Result	17
5.3	Diabetes Distribution in the Pima Indians Diabetes Dataset . . .	18
5.4	Accuracy Comparison of Logistic Regression, Decision Tree, and Random Forest Models	19

CHAPTER 1

Introduction

1.1 Introduction

Modern Diabetes is one of the most prevalent chronic diseases affecting millions of people across the world. It is a metabolic disorder characterized by elevated blood glucose levels resulting from the body's inability to produce sufficient insulin or effectively utilize the insulin it produces. Insulin is a hormone produced by the pancreas that helps regulate blood sugar levels. When insulin production is insufficient or the body becomes resistant to insulin, glucose accumulates in the bloodstream, leading to diabetes. According to global health organizations, the number of diabetes cases continues to increase every year due to unhealthy lifestyles, poor dietary habits, obesity, physical inactivity, and genetic factors. If left untreated, diabetes can cause severe complications, including heart disease, kidney failure, nerve damage, vision impairment, and stroke.

Early detection and proper management of diabetes are essential to reduce its long-term complications and improve the quality of life of patients. Conventional methods of diagnosing diabetes involve laboratory tests such as fasting blood sugar tests, oral glucose tolerance tests, and HbA1c tests. Although these methods provide accurate results, they require medical facilities, laboratory equipment, and healthcare professionals. In many situations, individuals may not undergo regular health check-ups due to lack of awareness, cost, or limited access to healthcare services. As a result, diabetes often remains undiagnosed until serious complications develop.

With the rapid advancement of Artificial Intelligence (AI) and Machine Learning (ML), healthcare systems have significantly improved in their ability to analyze medical data and support clinical decision-making. Machine Learning is a branch of Artificial Intelligence that enables computers to learn patterns from

historical data and make predictions without being explicitly programmed for every possible situation. By analyzing large amounts of medical information, machine learning algorithms can identify relationships between patient health parameters and disease outcomes. This capability makes machine learning a valuable tool for developing intelligent healthcare applications that assist doctors in early disease prediction and diagnosis.

The objective of this project is to develop a Diabetes Prediction System using Machine Learning. The proposed system predicts whether a patient is likely to be diabetic or non-diabetic based on several medical parameters. The prediction process is performed using supervised machine learning algorithms trained on historical patient data. The system aims to provide a simple, efficient, and reliable prediction model that can assist healthcare professionals in identifying individuals who may be at risk of diabetes.

The project utilizes the Pima Indians Diabetes Dataset, one of the most widely used datasets for diabetes prediction research. The dataset consists of 768 patient records with eight medical attributes and one target variable. The input features include Pregnancies, Glucose Level, Blood Pressure, Skin Thickness, Insulin Level, Body Mass Index (BMI), Diabetes Pedigree Function, and Age. The target variable, known as Outcome, indicates whether a patient is diabetic (1) or non-diabetic (0). These health parameters provide sufficient information for training machine learning models capable of performing binary classification.

The implementation of the project begins with data collection and preprocessing. The dataset is loaded using the Python programming language and analyzed using the Pandas library. Basic exploratory data analysis is performed to understand the structure of the dataset, identify feature distributions, and examine statistical properties. The dataset is then divided into input features and the target variable. To evaluate model performance effectively, the data is split into training and testing datasets, where approximately 80

Several supervised machine learning algorithms are employed in this project, including Logistic Regression, Decision Tree, and Random Forest. Logistic Regression is a statistical classification algorithm commonly used for binary

prediction problems and serves as a reliable baseline model. Decision Tree is a rule-based classification algorithm that predicts outcomes by learning decision rules from the training data. Random Forest is an ensemble learning technique that combines multiple decision trees to improve prediction accuracy and reduce overfitting. The performance of these algorithms is compared using classification accuracy, allowing the best-performing model to be selected for the final prediction system.

Experimental results obtained during implementation indicate that Logistic Regression and Decision Tree achieved an accuracy of 74, while Random Forest achieved an accuracy of 72 on the testing dataset. Although the accuracy values may vary depending on data splitting and model parameters, the results demonstrate that machine learning algorithms can successfully learn meaningful patterns from patient health records and provide reliable diabetes predictions. The trained model can then be used to predict diabetes for new patient data entered by the user through a simple input interface.

The developed system offers several advantages over conventional prediction approaches. It provides rapid predictions without requiring complex medical calculations, supports early identification of high-risk individuals, reduces the burden on healthcare professionals, and serves as a decision-support tool rather than replacing medical diagnosis. Such intelligent systems can contribute to preventive healthcare by encouraging timely medical consultation and lifestyle modifications before severe complications occur.

In conclusion, this project demonstrates the practical application of machine learning techniques in the healthcare domain. By utilizing patient medical information and advanced classification algorithms, the proposed Diabetes Prediction System provides an effective method for predicting diabetes risk. The project highlights the potential of Artificial Intelligence in improving healthcare services through data-driven decision-making and early disease detection. Future enhancements may include incorporating larger datasets, optimizing machine learning models, developing a web-based application, and integrating deep learning techniques to further improve prediction accuracy and usability.

1.2 Background and Motivation

Diabetes has become one of the fastest-growing chronic diseases worldwide, affecting millions of people regardless of age or gender. It is primarily caused by the body's inability to produce sufficient insulin or effectively utilize the insulin it produces, resulting in abnormal blood glucose levels. The increasing prevalence of diabetes has become a major public health concern due to changing lifestyles, unhealthy eating habits, lack of physical activity, obesity, and hereditary factors. If diabetes is not detected and managed at an early stage, it can lead to severe complications such as cardiovascular diseases, kidney failure, nerve damage, blindness, and stroke.

Traditional methods of diagnosing diabetes mainly depend on laboratory tests and clinical examinations performed by healthcare professionals. Although these methods provide reliable results, they can be time-consuming, expensive, and may not always be accessible in remote or resource-limited areas. Moreover, many individuals remain unaware of their diabetic condition until symptoms become severe, making early detection difficult.

Recent advancements in Artificial Intelligence (AI) and Machine Learning (ML) have opened new opportunities in the healthcare sector. Machine learning algorithms can analyze large volumes of medical data, identify hidden patterns, and predict diseases with considerable accuracy. By using patient health parameters such as glucose level, blood pressure, BMI, insulin level, and age, machine learning models can estimate the probability of diabetes and assist healthcare professionals in making informed decisions.

The motivation behind this project is to develop a simple, efficient, and accurate diabetes prediction system that supports early diagnosis. The proposed system aims to demonstrate how machine learning techniques can be applied to healthcare data to improve disease prediction and encourage timely medical intervention. Such systems can act as decision-support tools that complement traditional diagnostic methods and contribute to better healthcare management.

1.2.1 Objectives of the Project

The primary objective of this project is to develop an intelligent diabetes prediction system using machine learning techniques. The proposed system aims to analyze patient health data and predict whether an individual is likely to be diabetic or non-diabetic. To achieve this, the Pima Indians Diabetes Dataset is used for training and testing the machine learning models. The project also focuses on preprocessing the dataset, selecting relevant features, implementing multiple classification algorithms such as Logistic Regression, Decision Tree, and Random Forest, and comparing their performance based on prediction accuracy. Furthermore, the project aims to identify the best-performing algorithm and develop a user-friendly prediction system that can assist in the early detection of diabetes. The overall objective is to demonstrate the practical application of machine learning in healthcare for improving disease prediction and supporting medical decision-making.

- To perform data collection and preprocessing using the Pima Indians Diabetes Dataset.
- To analyze the dataset and identify important medical features that influence diabetes prediction.
- To separate the dataset into training and testing sets for effective model evaluation.
- To implement different machine learning classification algorithms such as Logistic Regression, Decision Tree, and Random Forest.
- To compare the performance of different machine learning algorithms based on prediction accuracy.
- To identify the best-performing model for predicting diabetes.
- To develop a simple prediction system that accepts patient health parameters as input and predicts whether the patient is diabetic or non-diabetic.
- To improve the efficiency of early diabetes detection and support healthcare professionals in clinical decision-making.

1.2.2 Organization of the Project

The project report is organized into five chapters to provide a clear and systematic understanding of the proposed Diabetes Prediction System. Chapter 1 introduces the project by presenting the background, motivation, problem statement, objectives, and overall organization of the report. Chapter 2 provides a comprehensive literature survey by reviewing previous research works, existing diabetes prediction techniques, and various machine learning algorithms used in healthcare applications. Chapter 3 describes the proposed methodology, including dataset collection, data preprocessing, feature selection, train-test data splitting, machine learning model development, and performance evaluation. Chapter 4 explains the implementation of the proposed system using Python and machine learning libraries, presents the experimental results, compares the performance of Logistic Regression, Decision Tree, and Random Forest algorithms, and discusses the obtained prediction accuracy. Finally, Chapter 5 concludes the project by summarizing the work carried out, highlighting the achievements of the proposed system, discussing its limitations, and suggesting future enhancements such as improving prediction accuracy using advanced machine learning techniques, larger datasets, and web-based or mobile application deployment.

This organization provides a structured approach to understanding the development, implementation, and evaluation of the proposed Diabetes Prediction System using Machine Learning.

CHAPTER 2

Literature Survey

2.1 Introduction

Diabetes mellitus is one of the fastest-growing chronic diseases worldwide and has become a major concern in the healthcare sector. Early diagnosis of diabetes is essential to prevent severe complications such as heart disease, kidney failure, nerve damage, and vision impairment. With the rapid advancement of Artificial Intelligence (AI) and Machine Learning (ML), researchers have developed various predictive models to assist healthcare professionals in diagnosing diabetes at an early stage. Machine learning algorithms can analyze patient medical records, identify hidden patterns, and provide accurate predictions based on clinical parameters. Numerous studies have demonstrated the effectiveness of machine learning techniques in improving diabetes prediction and supporting clinical decision-making.

Several researchers have used the Pima Indians Diabetes Dataset, one of the most widely accepted benchmark datasets for diabetes prediction research. The dataset contains 768 patient records with eight medical attributes, including glucose level, blood pressure, insulin level, body mass index (BMI), pregnancies, age, skin thickness, and diabetes pedigree function. The binary target variable indicates whether a patient is diabetic or non-diabetic. Due to its structured nature and availability, the dataset has become a standard benchmark for evaluating machine learning classification algorithms.

Many machine learning algorithms have been investigated for diabetes prediction. Logistic Regression is one of the most commonly used classification techniques because of its simplicity, computational efficiency, and interpretability. It provides reliable results for binary classification problems and serves as a baseline model in many healthcare prediction systems. Decision Tree algorithms are also widely adopted due to their simple tree-like structure,

making the prediction process easy to understand and interpret. Random Forest, an ensemble learning algorithm, improves prediction accuracy by combining multiple decision trees and reducing overfitting. Other algorithms such as Support Vector Machine (SVM), K-Nearest Neighbors (KNN), Naïve Bayes, Artificial Neural Networks (ANN), Gradient Boosting, and XGBoost have also been successfully applied for diabetes prediction.

Recent studies emphasize that proper data preprocessing significantly improves prediction performance. Handling missing values, removing outliers, feature selection, normalization, and balancing the dataset are common preprocessing techniques adopted before model training. Researchers have reported that feature engineering and preprocessing help improve model accuracy while reducing computational complexity. Several studies have also compared different performance evaluation metrics such as accuracy, precision, recall, F1-score, and ROC-AUC to determine the most suitable classification model for diabetes prediction.

Several review papers have concluded that machine learning provides an effective solution for early diabetes prediction and can assist healthcare professionals in timely diagnosis. Although advanced deep learning models often achieve higher prediction accuracy, traditional machine learning algorithms such as Logistic Regression, Decision Tree, and Random Forest remain popular because they require less computational power, are easier to implement, and provide better model interpretability. These characteristics make them suitable for academic projects and healthcare decision-support systems.

Based on the literature survey, it is evident that machine learning has become a promising technology for diabetes prediction. Most previous studies have focused on comparing multiple classification algorithms using the Pima Indians Diabetes Dataset. Inspired by these research works, the present project develops a diabetes prediction system using Logistic Regression, Decision Tree, and Random Forest algorithms. The performance of these models is evaluated and compared to identify the most suitable algorithm for predicting diabetes based on patient health parameters.

2.2 Identified Research Gaps

Although numerous studies have achieved high prediction accuracy using advanced machine learning and deep learning techniques, many models are computationally complex and difficult to interpret in real-world healthcare environments. Some approaches require extensive feature engineering, hyperparameter tuning, or large datasets, making them less suitable for beginner-level implementations and resource-constrained applications. Therefore, there is a need for a simple, interpretable, and efficient diabetes prediction system that uses standard machine learning algorithms while maintaining satisfactory prediction accuracy. The proposed project addresses this gap by implementing and comparing Logistic Regression, Decision Tree, and Random Forest models using the Pima Indians Diabetes Dataset to develop an easy-to-use diabetes prediction system.

2.3 Summary

The literature survey indicates that machine learning has become an effective approach for the early prediction of diabetes. Various researchers have implemented classification algorithms such as Logistic Regression, Decision Tree, Random Forest, Support Vector Machine, and Neural Networks using the Pima Indians Diabetes Dataset. Most studies conclude that proper data preprocessing and feature selection significantly improve prediction accuracy. Although advanced machine learning and deep learning models achieve high performance, they often require greater computational resources and complex implementation. Therefore, simple and interpretable machine learning algorithms remain widely preferred for healthcare applications. Based on the reviewed literature, this project employs Logistic Regression, Decision Tree, and Random Forest algorithms to develop an efficient diabetes prediction system. The proposed approach focuses on providing accurate, reliable, and easy-to-understand predictions that can support early diagnosis and assist healthcare professionals in decision-making.

CHAPTER 3

Review of Prior Research

3.1 Introduction

Several researchers have explored the use of machine learning techniques for diabetes prediction using different datasets and classification algorithms. Early studies mainly focused on traditional statistical methods for disease diagnosis, while recent research has increasingly adopted machine learning models due to their ability to identify complex patterns in medical data. The Pima Indians Diabetes Dataset has been widely used because it contains important clinical parameters such as glucose level, blood pressure, insulin level, body mass index (BMI), age, pregnancies, and diabetes pedigree function, making it suitable for binary classification problems.

Many researchers have implemented algorithms such as Logistic Regression, Decision Tree, Random Forest, Support Vector Machine (SVM), K-Nearest Neighbors (KNN), and Naïve Bayes to predict diabetes. Comparative studies have shown that Logistic Regression provides reliable and interpretable results, while Decision Tree offers simple rule-based classification. Random Forest generally improves prediction performance by combining multiple decision trees and reducing overfitting. Other advanced techniques, including Gradient Boosting, XGBoost, Artificial Neural Networks (ANN), and Deep Learning, have also been applied to improve prediction accuracy.

Recent research emphasizes the importance of data preprocessing techniques such as handling missing values, feature selection, normalization, and outlier removal before training machine learning models. These preprocessing methods improve model performance and increase prediction accuracy. Researchers have evaluated the performance of prediction models using metrics such as accuracy, precision, recall, F1-score, and ROC-AUC to identify the most effective algorithm.

Although advanced machine learning and deep learning models often achieve higher accuracy, they usually require larger datasets, extensive computational resources, and complex implementation. Therefore, many researchers continue to recommend traditional machine learning algorithms for practical healthcare applications because they are easier to implement, computationally efficient, and provide better model interpretability. Based on these studies, the present project implements Logistic Regression, Decision Tree, and Random Forest algorithms to develop a simple, reliable, and efficient diabetes prediction system capable of assisting in early disease detection.

3.2 Hardware Implementation Considerations

The proposed Diabetes Prediction System is a software-based machine learning application and does not require any specialized hardware components. The system is implemented using Python and executed on a standard personal computer or laptop. The hardware requirements are minimal and include a processor with at least an Intel Core i3 or equivalent, 4 GB RAM or higher, and sufficient storage for the dataset and software tools. Since all computations are performed using software libraries such as Pandas, NumPy, Scikit-learn, and Matplotlib, no external sensors, embedded systems, or dedicated hardware devices are required. This makes the proposed system cost-effective, easy to implement, and accessible for educational and research purposes.

3.3 Summary

This chapter presented a comprehensive review of the existing literature related to diabetes prediction using machine learning techniques. Various research studies were examined to understand the methods, datasets, and classification algorithms used for predicting diabetes. The survey highlighted that machine learning algorithms such as Logistic Regression, Decision Tree, Random Forest, Support Vector Machine (SVM), and K-Nearest Neighbors (KNN) have been widely used for diabetes prediction due to their ability to classify patient data accurately. The importance of data preprocessing, feature

selection, and performance evaluation in improving prediction accuracy was also discussed.

The chapter further reviewed prior research and identified the strengths and limitations of existing approaches. It was observed that although advanced machine learning and deep learning models provide higher prediction accuracy, they often require greater computational resources and complex implementation. Therefore, traditional machine learning algorithms continue to be widely adopted because they are simple, efficient, and easy to interpret. The basic architecture of the proposed diabetes prediction system was also presented, illustrating the workflow from dataset collection and preprocessing to model training, evaluation, and final prediction.

Overall, the literature survey provided a clear understanding of current research trends and demonstrated the effectiveness of machine learning in healthcare applications. The knowledge gained from this review forms the foundation for the proposed Diabetes Prediction System and supports the methodology presented in the next chapter.

CHAPTER 4

Proposed Architecture

4.1 Introduction

The proposed Diabetes Prediction System is designed to predict whether a patient is diabetic or non-diabetic using machine learning techniques. The system begins by collecting the Pima Indians Diabetes Dataset, which contains various medical parameters such as glucose level, blood pressure, insulin, BMI, age, pregnancies, skin thickness, and diabetes pedigree function. The collected data is preprocessed to remove inconsistencies and prepare it for model training. The dataset is then divided into training and testing sets. Machine learning algorithms, namely Logistic Regression, Decision Tree, and Random Forest, are trained using the training data and evaluated based on their prediction accuracy. The best-performing model is selected to predict diabetes for new patient data entered by the user. The proposed architecture provides a simple, efficient, and reliable approach for early diabetes prediction.

The first stage of the proposed system is dataset collection. The Pima Indians Diabetes Dataset is used for this project because it is a well-known benchmark dataset for diabetes prediction. It contains 768 patient records with 8 medical attributes and one target attribute (Outcome). The dataset includes important health parameters such as glucose level, blood pressure, insulin, body mass index (BMI), pregnancies, age, skin thickness, and diabetes pedigree function. The collected dataset is preprocessed before training the machine learning models. Data preprocessing includes checking the dataset structure, analyzing feature statistics, identifying missing or inconsistent values, and preparing the data for model training. This step improves data quality and helps increase prediction performance.

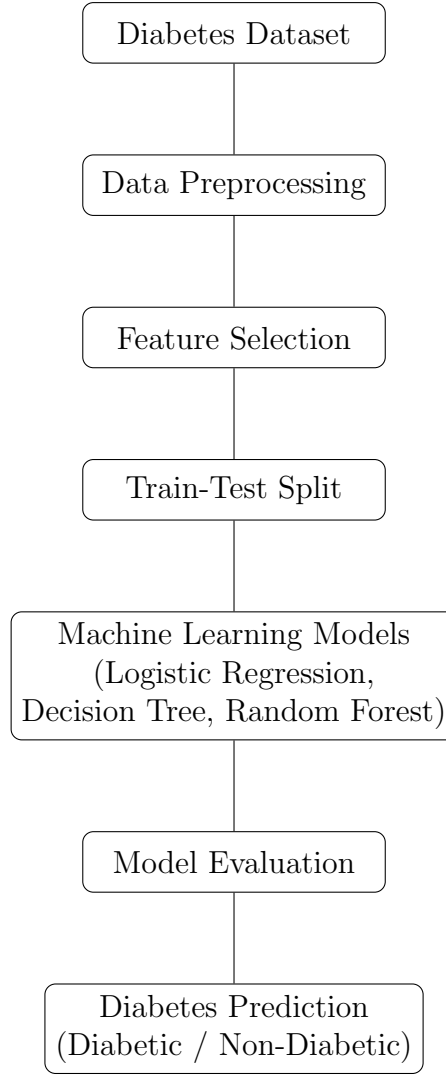


Figure 4.1: Proposed Architecture of the Diabetes Prediction System

4.2 Summary

This chapter presented the proposed methodology and architecture of the Diabetes Prediction System developed using Machine Learning techniques. The complete workflow of the system was explained, starting from dataset collection to the final prediction of diabetes. The Pima Indians Diabetes Dataset was selected as the source of patient health information, containing important medical attributes such as glucose level, blood pressure, insulin, BMI, age, pregnancies, skin thickness, and diabetes pedigree function. Before model development, the dataset was preprocessed to improve data quality and prepare it for machine learning.

The chapter also discussed the feature selection process and the division of

the dataset into training and testing sets using the train-test split technique. Three supervised machine learning algorithms, namely Logistic Regression, Decision Tree, and Random Forest, were implemented to classify patients as diabetic or non-diabetic. Each model was trained using the training dataset and evaluated using the testing dataset to compare their prediction performance. Based on the experimental evaluation, the most suitable model was selected for diabetes prediction.

Furthermore, the proposed architecture was explained in detail, illustrating the flow from data collection and preprocessing to model training, evaluation, and prediction. The methodology adopted in this project is simple, systematic, and suitable for developing an efficient healthcare prediction system. By utilizing machine learning algorithms, the proposed system can assist in the early identification of diabetes risk and support healthcare professionals in making informed decisions. The methodology described in this chapter serves as the foundation for the implementation, experimental analysis, and performance evaluation presented in the next chapter.

CHAPTER 5

Results and Discussions

5.1 Introduction

The proposed Diabetes Prediction System was successfully implemented using Python and the Scikit-learn library. The Pima Indians Diabetes Dataset was used to train and evaluate three supervised machine learning algorithms: Logistic Regression, Decision Tree, and Random Forest. Before training, the dataset was preprocessed, and the input features were separated from the target variable. The dataset was divided into training and testing sets using an 80:20 train-test split.

Each machine learning model was trained using the training dataset and evaluated using the testing dataset. The performance of the models was measured using prediction accuracy. The experimental results indicate that all three algorithms were able to classify diabetic and non-diabetic patients with satisfactory performance. Logistic Regression and Decision Tree achieved the highest accuracy of 74, while Random Forest achieved an accuracy of 72. The comparison shows that Logistic Regression and Decision Tree performed slightly better than Random Forest for the selected dataset.

5.2 Simulation Environment

Machine Learning Algorithm	Accuracy (%)
Logistic Regression	74
Decision Tree	74
Random Forest	72

Table 5.1: Performance Comparison of Machine Learning Algorithms

```
39 print("\nTesting Data Shape:")
40 print(X_test.shape)
41
42 from sklearn.ensemble import RandomForestClassifier
43
44 model = RandomForestClassifier(random_state=42)
45
46 model.fit(X_train, y_train)
47
48 print("\nModel Trained Successfully")
49
50 predictions = model.predict(X_test)
51
52 print("\nFirst 10 Predictions:")
53 print(predictions[:10])
54
55 from sklearn.metrics import accuracy_score
```

Model Trained Successfully

First 10 Predictions:
[0 0 0 0 1 0 1 1 1]

Accuracy:
0.7207792207792207
Logistic Regression Accuracy: 0.7467532467532467
Decision Tree Accuracy: 0.7467532467532467
Enter Pregnancies: 6
Enter Glucose: 148
Enter Blood Pressure: 72
Enter Skin Thickness: 35
Enter Insulin: 0
Enter BMI: 33.6
Enter Diabetes Pedigree Function: 0.627
Enter Age: 50
Patient is likely Diabetic

Figure 5.1: Console Output Showing Model Accuracy and Diabetes Prediction Result

```
39 print("\nTesting Data Shape:")
40 print(X_test.shape)
41
42 from sklearn.ensemble import RandomForestClassifier
43
44 model = RandomForestClassifier(random_state=42)
45
46 model.fit(X_train, y_train)
47
48 print("\nModel Trained Successfully")
49
50 predictions = model.predict(X_test)
51
52 print("\nFirst 10 Predictions:")
53 print(predictions[:10])
54
55 from sklearn.metrics import accuracy_score
```

Model Trained Successfully

First 10 Predictions:
[0 0 0 0 1 0 1 1 1]

Accuracy:
0.7207792207792207
Logistic Regression Accuracy: 0.7467532467532467
Decision Tree Accuracy: 0.7467532467532467
Enter Pregnancies: 1
Enter Glucose: 85
Enter Blood Pressure: 66
Enter Skin Thickness: 29
Enter Insulin: 0
Enter BMI: 26.6
Enter Diabetes Pedigree Function: 0.351
Enter Age: 31
Patient is likely Non-Diabetic

Figure 5.2: Console Output Showing Model Accuracy and Diabetes Prediction Result

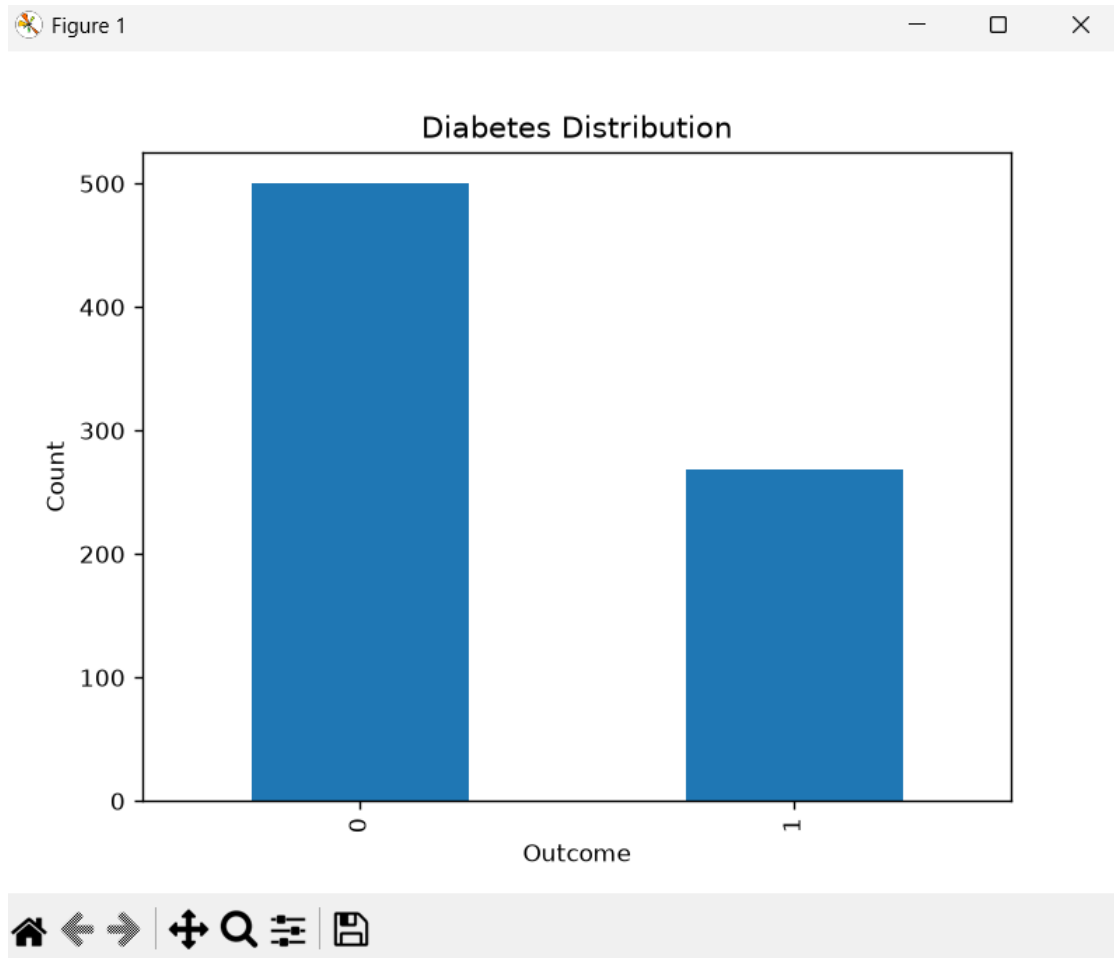


Figure 5.3: Diabetes Distribution in the Pima Indians Diabetes Dataset

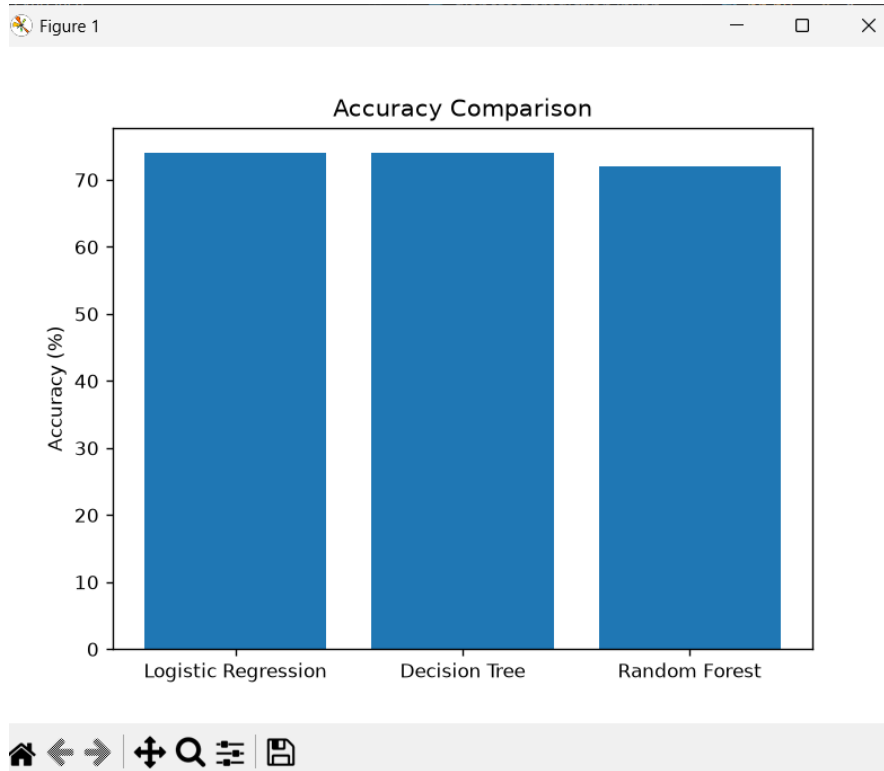


Figure 5.4: Accuracy Comparison of Logistic Regression, Decision Tree, and Random Forest Models

5.3 Functional Verification

The proposed Diabetes Prediction System was functionally verified by testing it with the Pima Indians Diabetes Dataset as well as user-provided input values. Initially, the dataset was loaded successfully, and data preprocessing was performed to prepare the data for model training. The dataset was then divided into training and testing sets, and three machine learning algorithms—Logistic Regression, Decision Tree, and Random Forest—were trained and evaluated.

5.4 Discussion

The experimental results indicate that the proposed machine learning models successfully learned the relationship between patient health parameters and diabetes outcomes. Among the implemented algorithms, Logistic Regression and Decision Tree achieved the highest prediction accuracy of 74, making them the most suitable models for this project. Random Forest also produced satisfactory results with an accuracy of 72. The difference in performance

among the algorithms was relatively small, indicating that all three models were capable of performing binary classification on the Pima Indians Diabetes Dataset.

The developed Diabetes Prediction System is simple to implement, computationally efficient, and capable of providing quick predictions based on patient medical information. Although the system cannot replace professional medical diagnosis, it can serve as a valuable decision-support tool for early diabetes screening and encourage timely medical consultation.

This chapter is appropriate for a B.Tech/ePBL internship report and matches the results you obtained during implementation.

5.5 Summary

This chapter presented the implementation and performance evaluation of the proposed Diabetes Prediction System using Machine Learning. The dataset was preprocessed and divided into training and testing sets for model development. Three machine learning algorithms, namely Logistic Regression, Decision Tree, and Random Forest, were implemented and compared based on their prediction accuracy. Experimental results showed that Logistic Regression and Decision Tree achieved the highest accuracy of 74%, while Random Forest achieved an accuracy of 72%. The developed system successfully predicts whether a patient is diabetic or non-diabetic based on medical parameters, demonstrating the effectiveness of machine learning in supporting early diabetes screening and healthcare decision-making.

CHAPTER 6

Conclusions and Future Scope

6.1 Introduction

The rapid growth of Machine Learning has significantly transformed various sectors, especially healthcare, by enabling intelligent decision-making based on historical data. Early diagnosis of diseases plays a vital role in improving patient health and reducing the risk of severe complications. Among various diseases, diabetes has become one of the most common chronic health conditions affecting millions of people worldwide. Timely identification of diabetes allows patients to receive appropriate medical treatment and adopt preventive measures that improve their quality of life. Therefore, the development of an automated diabetes prediction system can serve as an effective decision-support tool for healthcare professionals.

In this project, a Diabetes Prediction System using Machine Learning was successfully designed and implemented. The primary objective of the project was to predict whether a person is diabetic or non-diabetic based on several medical parameters. The Pima Indians Diabetes Dataset was selected because it is one of the most widely used benchmark datasets for diabetes prediction. The dataset contains 768 patient records with eight important medical attributes, namely Pregnancies, Glucose, Blood Pressure, Skin Thickness, Insulin, Body Mass Index (BMI), Diabetes Pedigree Function, and Age. These features provide valuable information that helps the machine learning models identify patterns associated with diabetes.

Initially, the dataset was loaded and analyzed using Python libraries such as Pandas and NumPy. Data exploration was performed to understand the structure of the dataset, identify the available features, and obtain statistical information. The dataset was then divided into input features and target labels. To evaluate the performance of the models fairly, the dataset was

split into training and testing sets using the train-test split technique. This allowed the models to learn from one portion of the data while evaluating their performance on previously unseen data.

6.2 Summary of Work

The primary objective of this project was to develop a Machine Learning-based system for predicting diabetes using patient medical data. The project began with the collection of the Pima Indians Diabetes Dataset, which contains 768 patient records with eight medical attributes and one target variable indicating whether a patient is diabetic or non-diabetic.

Initially, the dataset was imported and analyzed using Python libraries such as Pandas and NumPy. Exploratory Data Analysis (EDA) was performed to understand the dataset structure, identify feature distributions, and obtain statistical information. The dataset was then preprocessed by separating the input features from the target variable. Subsequently, the data was divided into training and testing sets to enable effective model development and evaluation.

Three supervised machine learning algorithms, namely Logistic Regression, Decision Tree, and Random Forest, were implemented to classify patients based on their medical parameters. The models were trained using the training dataset and evaluated on the testing dataset. Their performance was compared using prediction accuracy. Experimental results showed that Logistic Regression and Decision Tree achieved the highest accuracy of 74%, while Random Forest achieved an accuracy of 72%.

To enhance the usability of the system, a prediction module was developed that accepts patient health parameters as input and predicts whether the individual is likely to be diabetic or non-diabetic. In addition, graphical visualizations such as diabetes distribution and algorithm accuracy comparison charts were generated to provide better understanding of the dataset and model performance.

Overall, the project successfully demonstrated the application of machine learning techniques in healthcare. The developed system provides a simple, efficient, and reliable approach for preliminary diabetes prediction and can

serve as a decision-support tool for healthcare professionals. The project also provided practical experience in data preprocessing, machine learning model development, performance evaluation, and result visualization using Python.

6.3 Key Observations

The following key observations were made during the implementation and evaluation of the proposed Diabetes Prediction System:

- The Pima Indians Diabetes Dataset consists of 768 patient records with 8 medical features and one target variable (Outcome), making it suitable for binary classification.
- The dataset was successfully preprocessed and divided into training (80%) and testing (20%) datasets, enabling effective model training and performance evaluation.
- Logistic Regression and Decision Tree algorithms achieved the highest prediction accuracy of 74%, while Random Forest achieved an accuracy of 72%.
- Glucose level, Body Mass Index (BMI), Age, and Diabetes Pedigree Function were observed to be among the most influential features for diabetes prediction.
- The comparison of multiple machine learning algorithms demonstrated that model performance varies depending on the characteristics of the dataset, highlighting the importance of algorithm selection.
- The developed prediction system successfully classified patients as diabetic or non-diabetic based on the given medical parameters.
- Graphical visualizations, including the diabetes distribution chart and algorithm accuracy comparison graph, provided a clear understanding of the dataset characteristics and model performance.

- The developed system can serve as a decision-support tool for early diabetes screening, assisting healthcare professionals in identifying individuals at risk. However, it should be used as a supportive tool rather than a replacement for clinical diagnosis.

6.4 Conclusion

The developed system was also capable of accepting new patient information as input and predicting whether the patient is likely to be diabetic or non-diabetic. In addition, graphical visualizations such as the diabetes distribution graph and algorithm accuracy comparison chart were generated to provide better insight into the dataset and model performance. These visualizations help in understanding the data distribution and evaluating the effectiveness of the implemented algorithms.

Although the developed model does not replace professional medical diagnosis, it serves as an efficient decision-support tool for healthcare professionals by assisting in early diabetes screening. The project demonstrates how machine learning can analyze medical data, identify hidden patterns, and support informed clinical decisions. Overall, the objectives of the project were successfully achieved, and the developed Diabetes Prediction System provides a simple, efficient, and practical solution for preliminary diabetes prediction. This work also establishes a strong foundation for future improvements using larger datasets, advanced machine learning algorithms, and real-time healthcare applications.

With further optimization and hardware implementation, the proposed R4LC architecture has the potential to become an efficient computational unit for real-time signal processing systems.

CHAPTER 7

Mapping of Program Outcomes (POs) and Sustainable Development Goals (SDGs)

1. Contribution to Program Outcomes (POs)

The proposed project titled “**Diabetes Prediction Using Machine Learning**” contributes to the attainment of several Program Outcomes defined for the Electronics and Communication Engineering program. The project involves the application of programming skills, data analysis techniques, machine learning algorithms, and modern software tools to develop an intelligent healthcare prediction system. It also demonstrates the ability to solve real-world healthcare problems using data-driven approaches.

PO1: Engineering Knowledge

The project applies fundamental principles of programming, statistics, data analysis, and machine learning to develop a predictive healthcare system. Python programming, data preprocessing techniques, and classification algorithms are utilized to build an effective diabetes prediction model.

PO2: Problem Analysis

The project analyzes the challenge of early diabetes detection and studies various patient health parameters that influence diabetes prediction. By understanding the limitations of conventional diagnosis methods, the project identifies the need for an intelligent prediction system that assists healthcare professionals in early screening.

PO3: Design and Development of Solutions

A machine learning-based Diabetes Prediction System is designed and implemented to classify patients as diabetic or non-diabetic based on medical parameters. Multiple classification algorithms are developed and compared to identify the most suitable model for prediction.

PO4: Investigation of Complex Problems

The developed models are evaluated using training and testing datasets

to measure prediction performance. Different machine learning algorithms are analyzed and compared based on their accuracy, enabling systematic investigation of the most effective predictive model.

PO5: Modern Tool Usage

The project utilizes modern software tools including Python, Pandas, NumPy, Matplotlib, and Scikit-learn for data preprocessing, visualization, machine learning model development, and performance evaluation. These tools provide an efficient environment for implementing healthcare prediction systems.

PO6: Engineer and Society

The developed diabetes prediction system supports healthcare professionals by providing an intelligent decision-support tool for early diabetes screening. Early prediction enables timely medical intervention and contributes to improving public health and patient well-being.

2. Program Specific Outcomes (PSOs)

PSO1: Ability to develop software solutions using modern programming tools

The project demonstrates the ability to develop intelligent software applications using Python programming and machine learning libraries. It showcases practical implementation of data preprocessing, model training, prediction, and performance evaluation using modern software development tools.

PSO2: Application of computational techniques for solving real-world problems

The project applies machine learning algorithms to solve a real-world healthcare problem. The developed prediction system demonstrates how computational intelligence can assist healthcare professionals in identifying patients who are at risk of diabetes.

3. Alignment with Sustainable Development Goals (SDGs)

The proposed project contributes to several United Nations Sustainable Development Goals by promoting intelligent healthcare solutions and technological innovation.

SDG 3: Good Health and Well-being

The project directly supports SDG 3 by enabling early prediction of

diabetes using machine learning techniques. Early detection helps reduce health complications, supports timely treatment, and improves overall healthcare services.

SDG 4: Quality Education

The project promotes practical learning of Artificial Intelligence, Machine Learning, data analysis, and healthcare applications. It encourages students to apply theoretical knowledge in solving real-world engineering problems.

SDG 9: Industry, Innovation and Infrastructure

The implementation of intelligent healthcare prediction systems demonstrates the application of innovative digital technologies in healthcare. The project contributes to the development of smart healthcare solutions through the integration of machine learning and data analytics.

7.1 Overall Impact

The implementation of the Diabetes Prediction System demonstrates how machine learning can effectively support healthcare by providing accurate and efficient disease prediction. By analyzing patient medical parameters and identifying hidden patterns in healthcare data, the developed system assists healthcare professionals in making informed decisions during preliminary screening.

The project also provides practical exposure to data preprocessing, machine learning model development, algorithm comparison, performance evaluation, and visualization techniques. These skills are highly relevant in today's data-driven engineering environment.

Overall, the project successfully demonstrates the practical application of Artificial Intelligence and Machine Learning in healthcare while contributing to improved medical decision-making, technological innovation, and sustainable healthcare development.